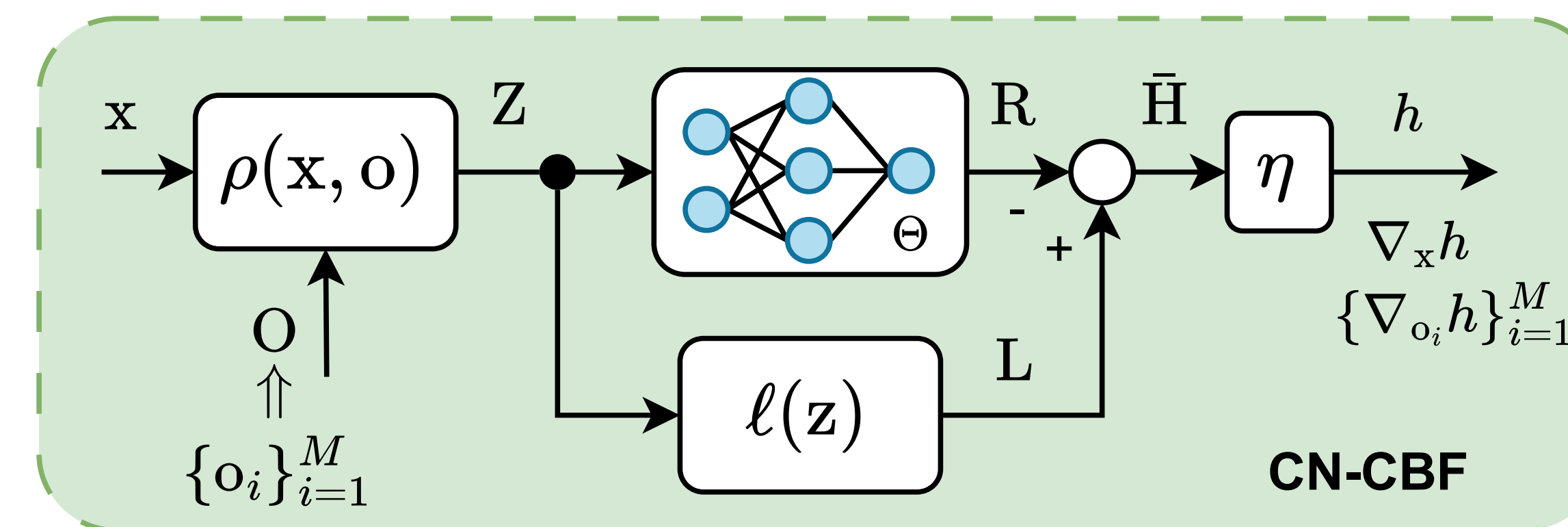


## MOTIVATION

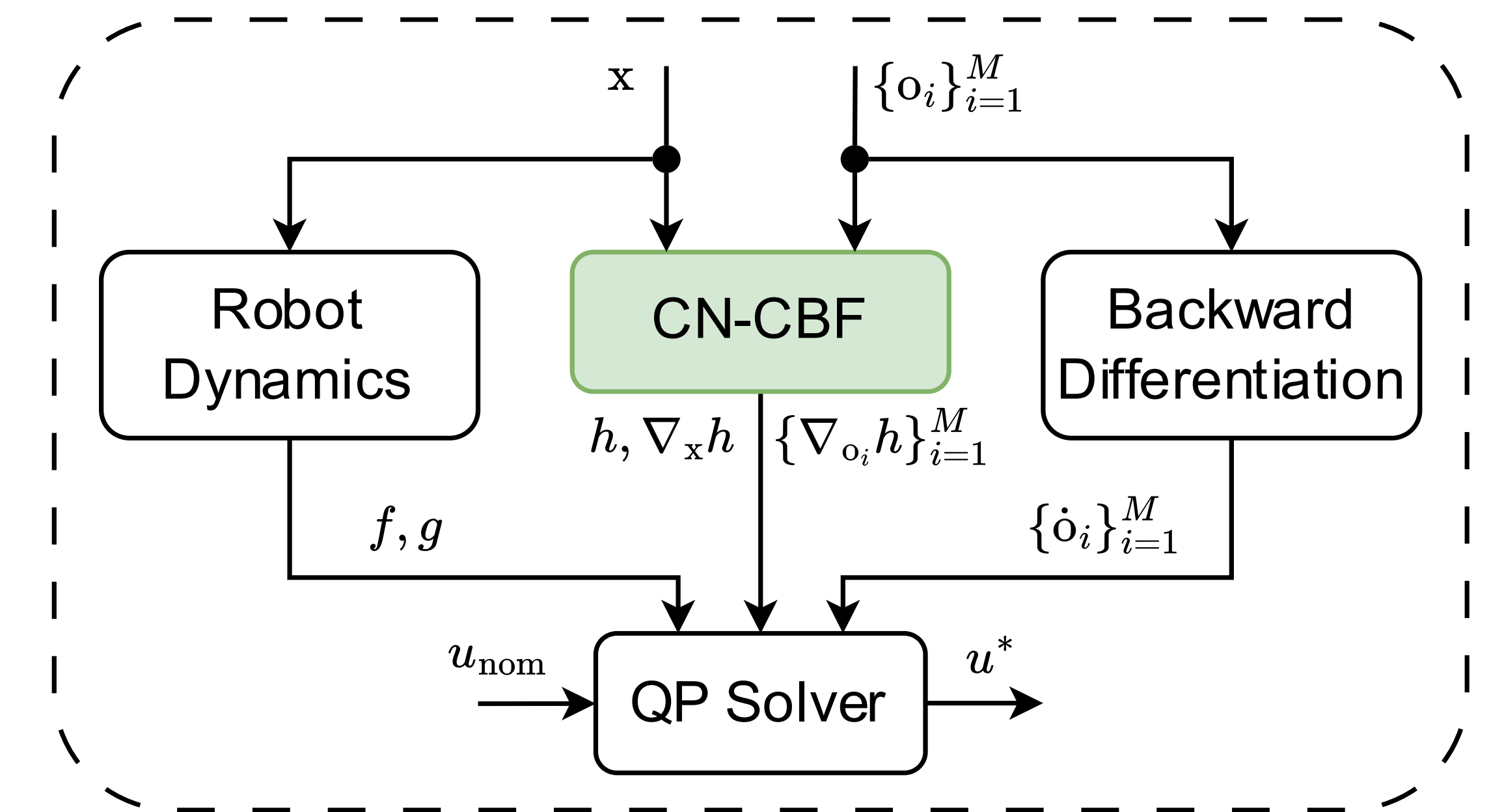
- Safe navigation in dynamic and uncertain environments remains a major challenge for autonomous robots.
- Control barrier function (CBF) safety filters enable real-time collision avoidance, but suitable CBFs are difficult to design.
- Hamilton-Jacobi (HJ) reachability provides near-optimal safe sets, but grid-based solutions are inefficient for online deployment.
- Existing learning-based methods often struggle to efficiently handle a varying number of moving obstacles.

## METHODOLOGY

- Compute the safe set for a single robot-obstacle pair offline using HJ reachability on the relative dynamics.
- Train a residual neural model to approximate the HJ value function while keeping failure states outside the learned safe set.
- Smoothly aggregate a batch of individual CBF values for any number of obstacles into a single composite CBF.

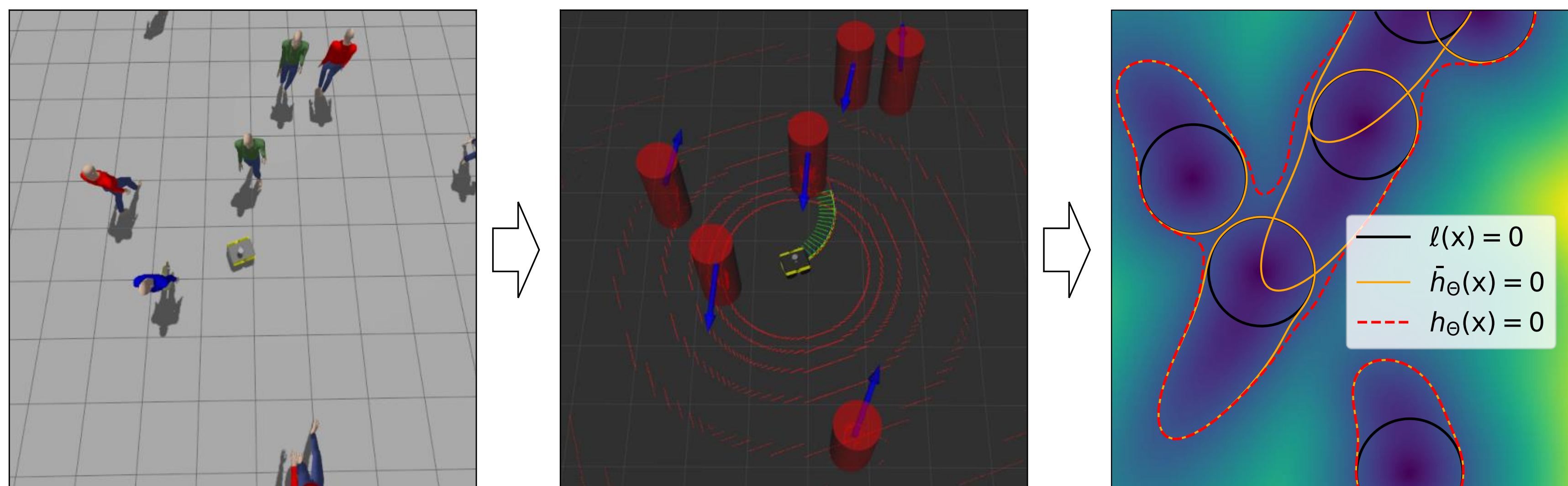


- Integrate the composite CBF into a CBF-QP safety filter that minimally modifies the nominal control.

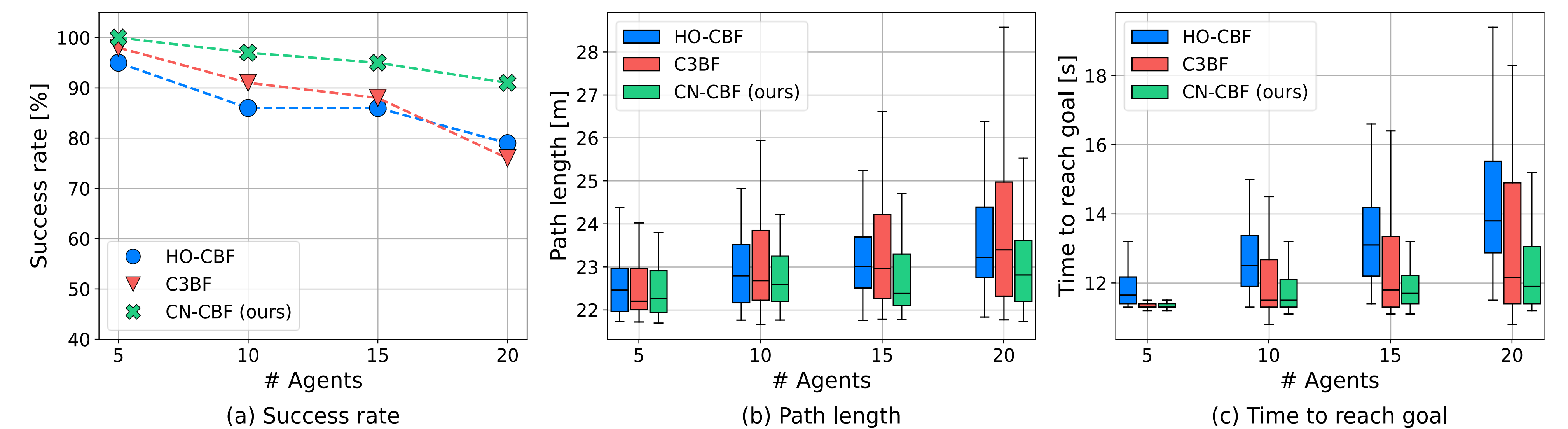


## RESULTS

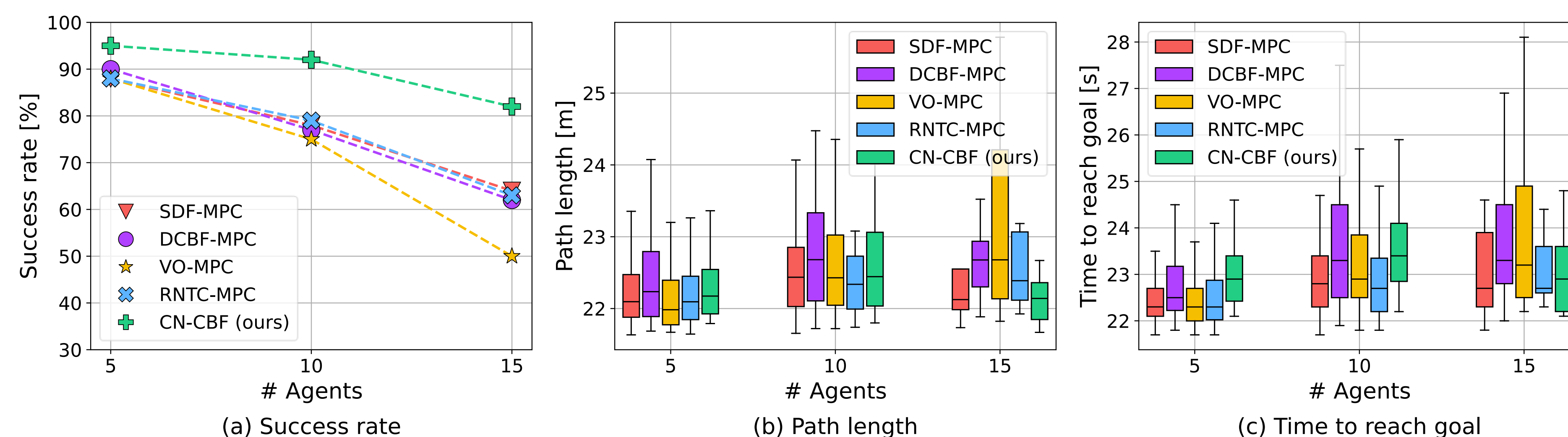
- A sample scene from the simulation experiments with the ground mobile robot:



- Evaluation against distance- and velocity-based CBFs on the quadcopter model:



- Evaluation against different MPC-based motion planners on the ground robot:



- Results from a real-world experiment with Crazyflie drones:

